

Solving Non-homogenous Recurrence Relations

Today we continue develop a second method for solving non-homogenous recurrence relations, using the auxiliary equation.

Class Session Goals:

- 1 Practice creating recurrence relations
- 2 Demonstrate elimination non-homogenous terms in recurrence

Some Non-homogenous Recurrences

von Neuman Neighborhoods (A001844)

$$V_n = V_{n-1} + 4(n-1) \text{ with } V_0 = 1$$

Tower of Hanoi (A001787)

$$H_n = 2H_{n-1} + 1 \text{ with } H_1 = 1$$

$2 \times 2n$ binary matrices with $n+1$ 1s (A000124)

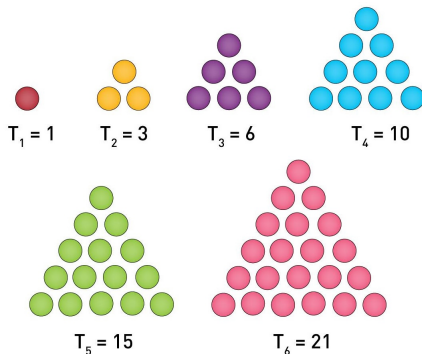
$$a_n = 2a_{n-1} + 2^{n-1} \text{ with } a_1 = 1$$

Lazy Caterer (A000124)

$$c_n = c_{n-1} + n \text{ with } c_0 = 1$$

Nested Triangular Numbers

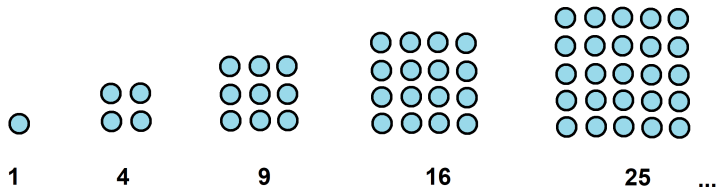
Figurative numbers with shapes with 3 sides.



$$T_n = T_{n-1} + n \text{ with } T_1 = 1 \text{ (A000217).}$$

Nested Square Numbers

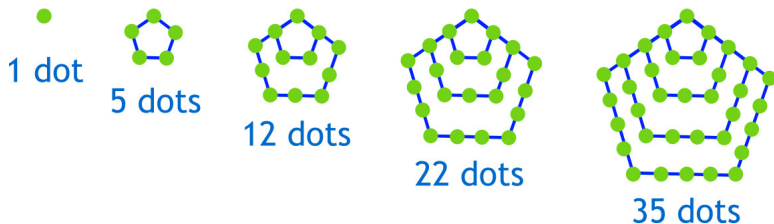
Figurative numbers with shapes with 4 sides.



$$S_n = S_{n-1} + 2n - 1 \text{ with } S_1 = 1 \text{ (A000290).}$$

Nested Pentagonal Numbers

Figurative numbers with shapes with 5 sides.



$$P_n = P_{n-1} + 3n - 2 \text{ with } P_1 = 1 \text{ (A000326).}$$

Calculation Demonstration

Exponential non-homogeneous terms with Towers of Hanoi recursion:

$$H_n = H_{n-1} + 2^{n-1} \text{ with } H_1 = 1$$

Polynomial non-homogeneous terms with Triangular numbers recursion:

$$T_n = T_{n-1} + n \text{ with } T_1 = 1$$

Now you try it

Eliminate the non-homogenous term for the following recurrence relations:

Alternate Tower of Hanoi (A001787)

$$H_n = 2H_{n-1} + 1 \text{ with } H_1 = 1$$

$2 \times 2n$ binary matrices with $n + 1$ 1s (A000124)

$$a_n = 2a_{n-1} + 2^{n-1} \text{ with } a_1 = 1$$

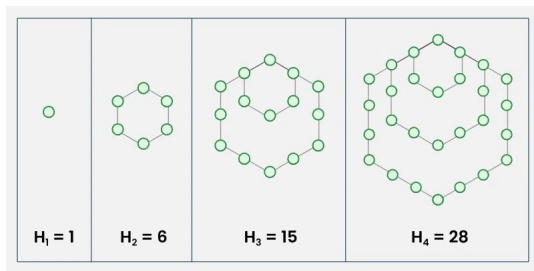
von Neuman Neighborhoods (A001844)

$$V_n = V_{n-1} + 4(n-1) \text{ with } V_0 = 1$$

Square Numbers (A000290)

$$S_n = S_{n-1} + 2n - 1 \text{ with } S_1 = 1$$

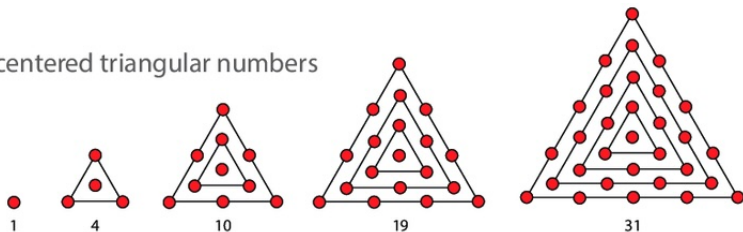
Nested Hexagonal Numbers



Let H_n be the number of dots in the n -th nested hexagon.

Centered Triangular Numbers

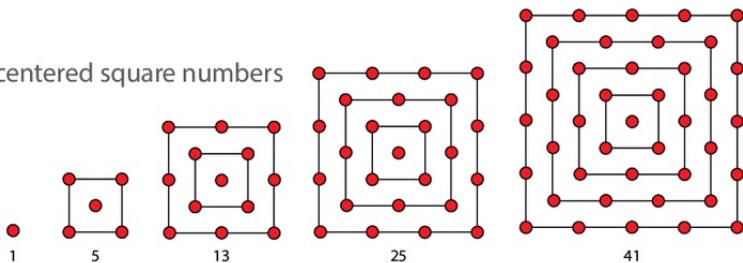
centered triangular numbers



Let $\mathcal{T}_n$ be the number of dots in the n -th center nested triangle.

Centered Square Numbers

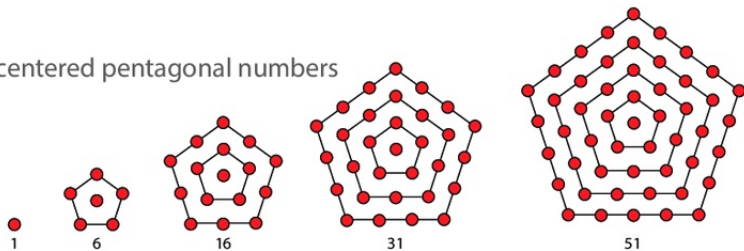
centered square numbers



Let S_n is the number of dots in the n -th center nested square .

Centered Pentagonal Numbers

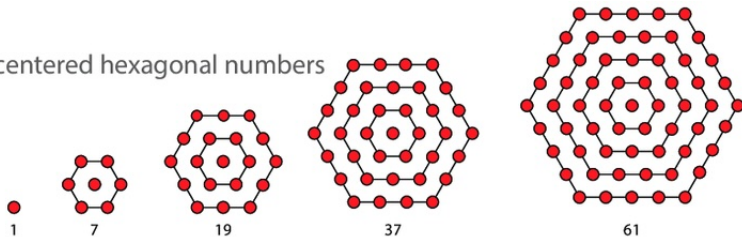
centered pentagonal numbers



Let $\mathcal{P}_n$ be the number of dots in the n -th center nested pentagon. Can you express $\mathcal{P}_n$ as a recurrence relation? What's its generating function? Can you find a closed form solution?

Centered Hexagonal Numbers

centered hexagonal numbers



Let $\mathcal{H}_n$ be the number of dots in the n -th center nested hexagon. Can you express $\mathcal{H}_n$ as a recurrence relation? What's its generating function? Can you find a closed form solution?

Activity

$$H_n = (1, 6, 15, 28, 45, \dots)$$

$$\mathcal{T}_n = (1, 4, 10, 19, 31, \dots)$$

$$\mathcal{S}_n = (1, 5, 13, 25, 41, \dots)$$

$$\mathcal{P}_n = (1, 6, 16, 31, 51, \dots)$$

$$\mathcal{H}_n = (1, 7, 19, 37, 61, \dots)$$

Choose one of these new figurative numbers and answer the following questions: What is the recurrence relationship? What is the recurrence relationship if you convert it into a homogeneous recurrence? What is the close form solution?